NetCore Systems Ltd. Network Implementation Report

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Summary

I built a segmented office network for NetCore Systems Ltd. in Cisco Packet Tracer that meets all four management requirements. IT, Sales, and HR each sit on their own subnet behind one Cisco 2911 router. The IT server issues addresses by DHCP and hosts the company web page and FTP service. An access control list on the router stops Sales from pinging the IT network while leaving the server's HTTP and FTP services available to Sales.

This report follows the work in the order it was carried out, in twelve steps, with the screenshot evidence placed under each step.

Problem and requirements

NetCore Systems Ltd. had no structured local area network for its three departments. There was no central addressing, no centrally managed file or web service, and no control over traffic between sensitive departments.

As the Network Technician, I was asked to:

- Create a separate network segment for each department.
- Assign IP addresses appropriately, using DHCP for the PCs.
- Provide HTTP and FTP services from one IT server.
- Test connectivity between departments with ping.
- Apply a basic ACL so that Sales cannot ping IT but can still use the server's services.

Network design

The design uses one /24 subnet per department, with the router interface as each subnet's default gateway.

Department	Network	Gateway (router interface)	DHCP range	Devices
IT	192.168.10.0/24	192.168.10.1 (G0/0)	From 192.168.10.5	PC0, PC1, Server0, Switch2
Sales	192.168.20.0/24	192.168.20.1 (G0/1)	From 192.168.20.5	PC4, PC5, Switch1
HR	192.168.30.0/24	192.168.30.1 (G0/2)	From 192.168.30.5	PC2, PC3, Switch0

The IT server (Server0) has a static address 192.168.10.2 and provides DHCP, HTTP, and FTP. The devices kept their default Packet Tracer names, so this report uses those names.

Building the network (steps 1 to 6)

Step 1: Create the topology

I added one Cisco 2911 router, three Cisco 2960 switches, six PCs, and one server, then connected them with copper straight-through cables. Router G0/0 goes to the IT switch, G0/1 to the Sales switch, and G0/2 to the HR switch. Each switch connects to its department's two PCs, and the IT switch also connects to the server. All link lights are green in the finished topology.

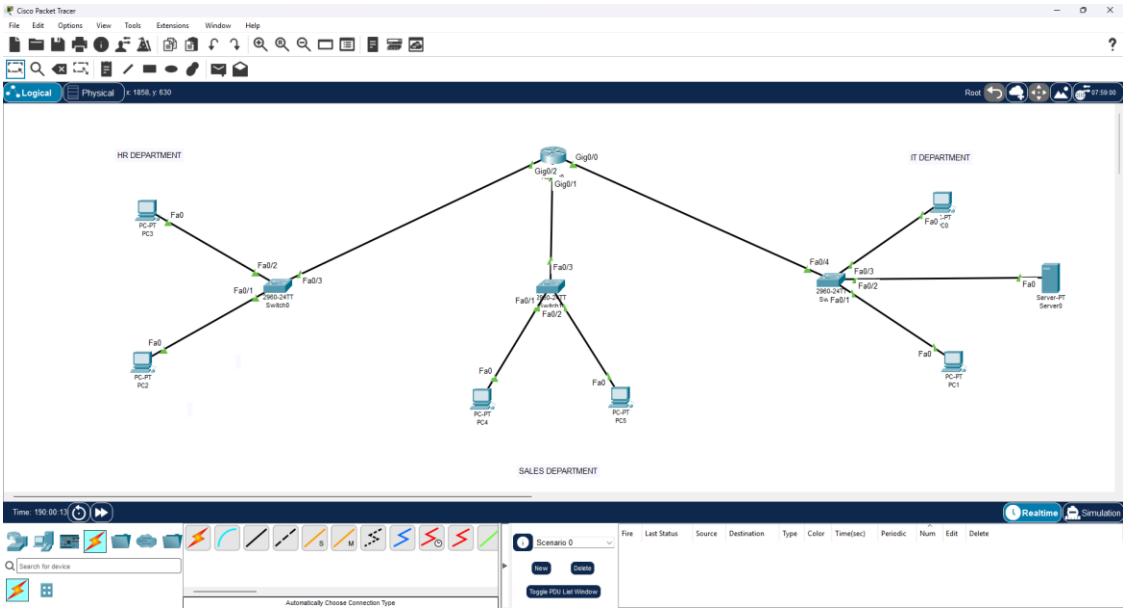


Figure 1. Completed topology with interface labels shown.

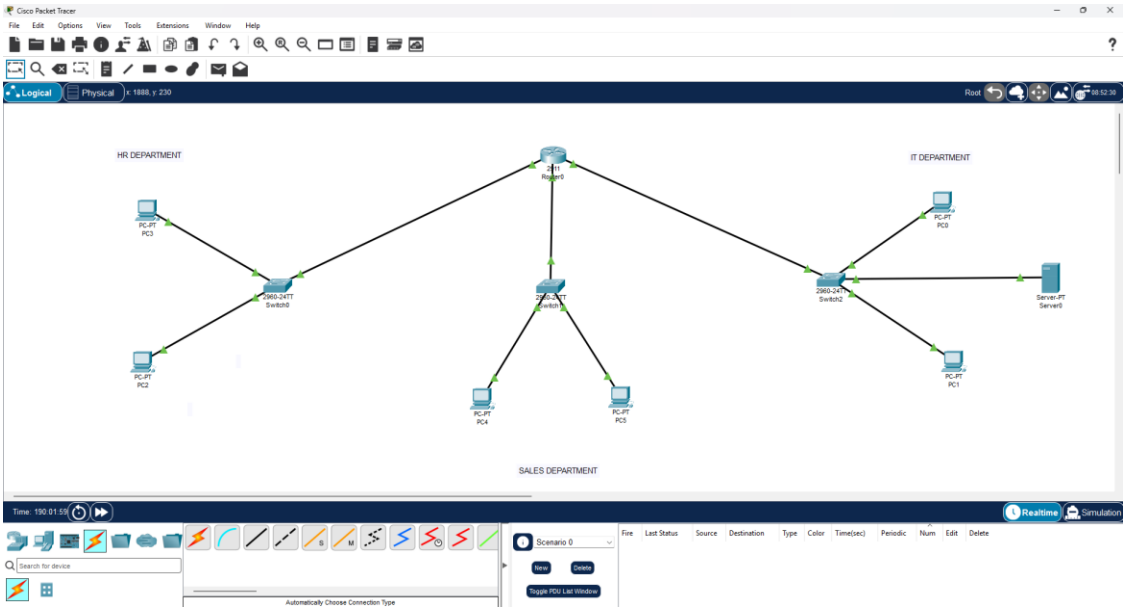


Figure 2. Completed topology with device names shown.

Step 2: Give the server a static address

The server needs a fixed address because clients must always find its services in the same place. On Server0, under Desktop > IP Configuration, I set:

IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

No screenshot was captured for this step. The address is confirmed later, where every PC reaches the server at 192.168.10.2 (Figures 14 and 15).

Step 3: Configure the router interfaces

Each router interface becomes the default gateway for one department. In the router CLI, I entered:

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit
interface gigabitEthernet 0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit
interface gigabitEthernet 0/2
 ip address 192.168.30.1 255.255.255.0
 no shutdown
 exit
```

```
Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
```

Figure 3. Addressing the three router interfaces and bringing them up.

I then ran

```
 show ip interface brief
```

to confirm the result. All three interfaces show the correct address with Status up and Protocol up.

```
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     192.168.30.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

Figure 4. All three GigabitEthernet interfaces are up/up with the planned addresses.

Step 4: Configure DHCP

I ran DHCP centrally on the IT server, in line with the goal of centralising network services. On Server0, under Services > DHCP, I turned the service on and created one pool per department. Each pool hands out its own department's gateway, the server as DNS, and addresses starting at .5.

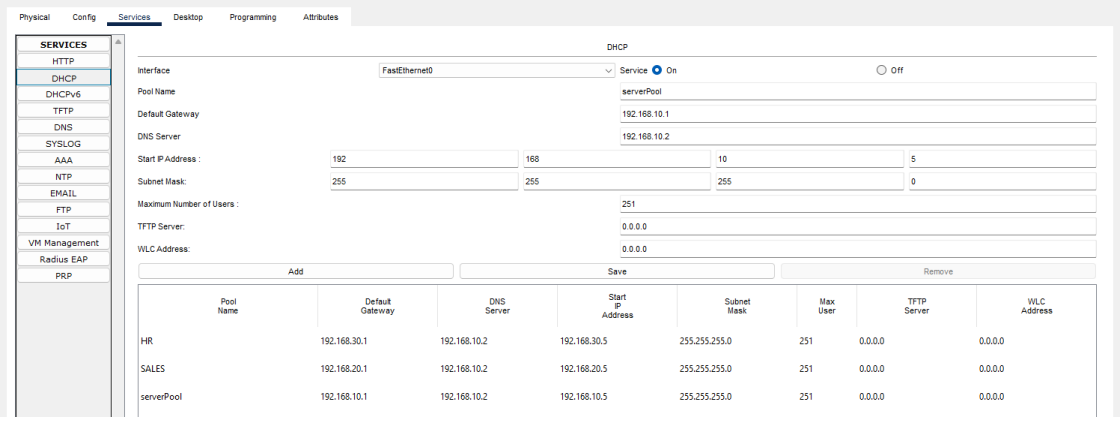


Figure 5. DHCP pools on Server0: serverPool for IT (192.168.10.0), SALES (192.168.20.0), and HR (192.168.30.0).

Sales and HR are on different subnets from the server, and routers do not forward DHCP requests, which are broadcasts. I therefore added a helper address on the Sales and HR router interfaces, so the router relays those requests to the server:

```
interface gigabitEthernet 0/1
 ip helper-address 192.168.10.2
 exit
interface gigabitEthernet 0/2
```

ip helper-address 192.168.10.2
exit

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip helper-address 192.168.10.2
Router(config-if)#exit
Router(config)#interface g0/2
Router(config-if)#ip helper-address 192.168.10.2
Router(config-if)#exit
```

Figure 6. DHCP relay configured on the Sales (G0/1) and HR (G0/2) interfaces.

Finally, I set every PC to DHCP under Desktop > IP Configuration. Each PC received an address in its own department's range.

Department	PC	Address received	Figure
IT	PC0	192.168.10.7	7
IT	PC1	192.168.10.5	8
Sales	PC4	192.168.20.5	9
Sales	PC5	192.168.20.7	10
HR	PC2	192.168.30.7	11
HR	PC3	192.168.30.6	12

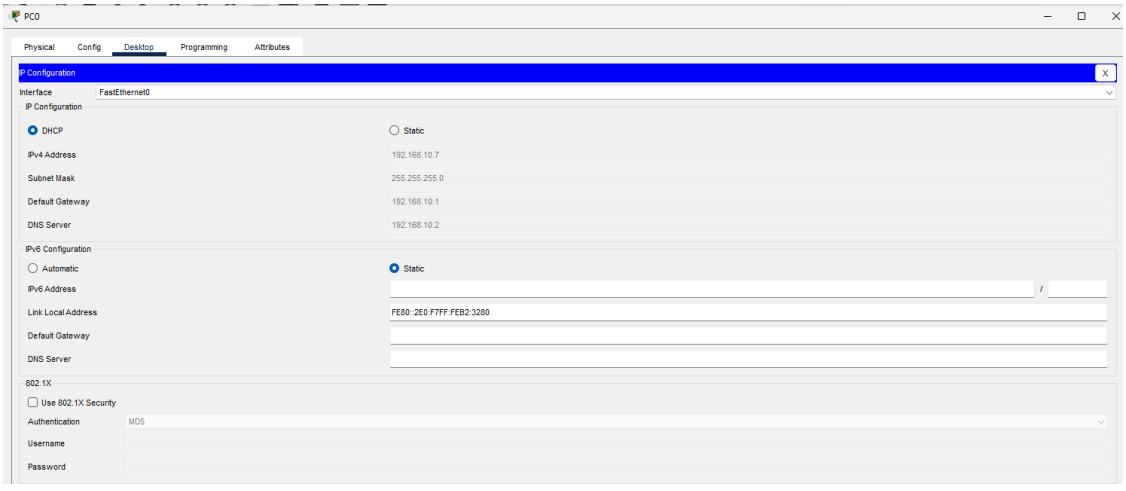


Figure 7. IT, PC0.

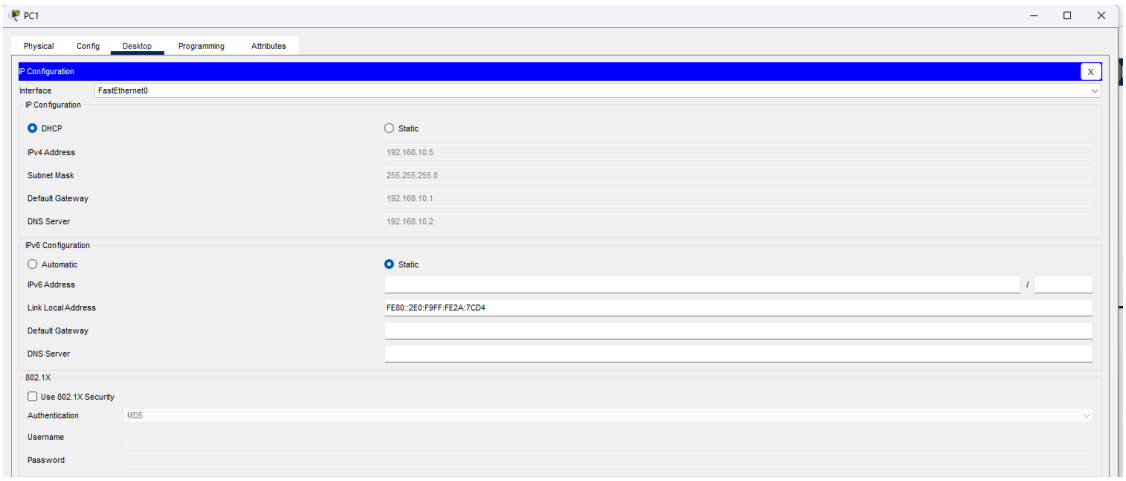


Figure 8. IT, PC1.

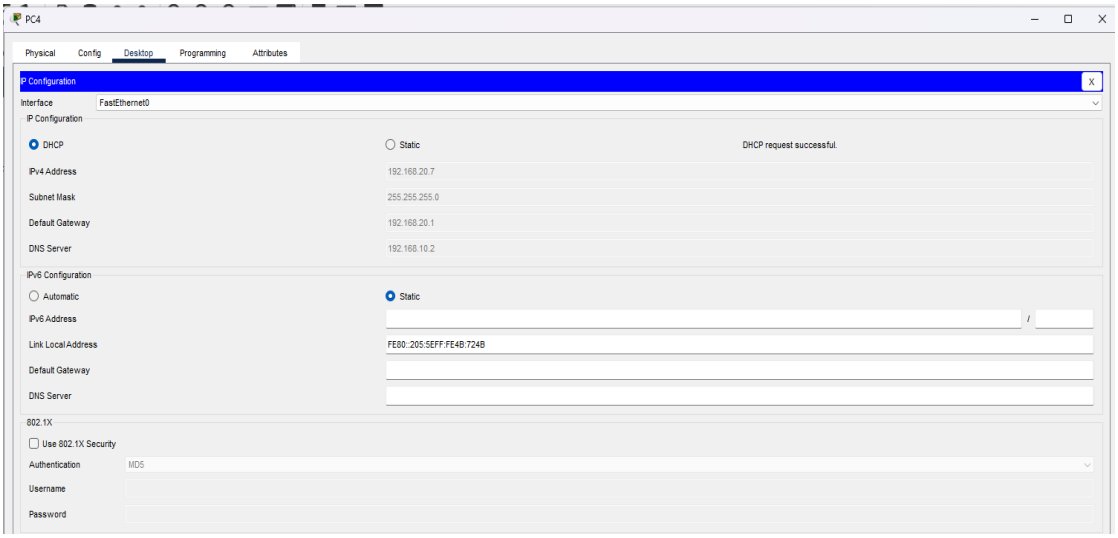


Figure 9. Sales, PC4.

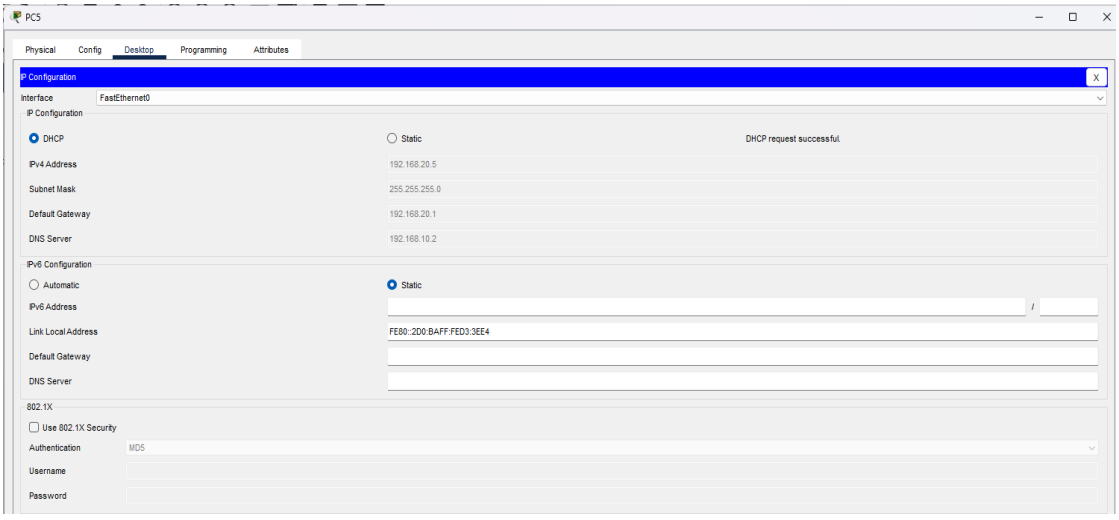


Figure 10. Sales, PC5.

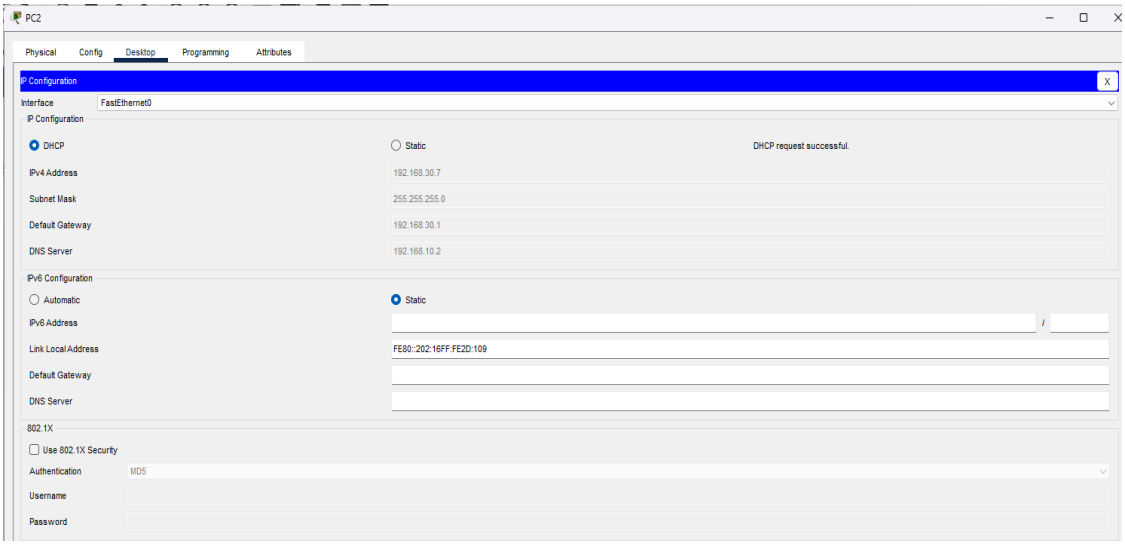


Figure 11. HR, PC2.

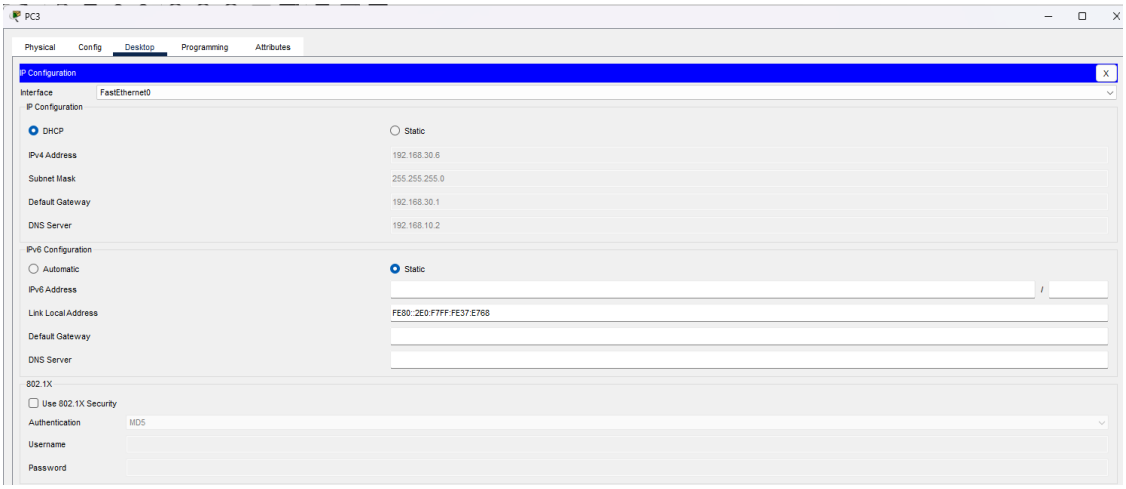


Figure 12. HR, PC3.

Step 5: Enable HTTP on the server

On Server0, under Services > HTTP, I turned HTTP on and edited the default page to read “NetCore Systems Ltd.” and “Internal Company Server”. No screenshot was captured of the service page. The edited page is visible when it loads in the browser test (Figure 14).

Step 6: Enable FTP on the server

On Server0, under Services > FTP, I turned FTP on and created the user technova with full permissions (read, write, delete, rename, list).

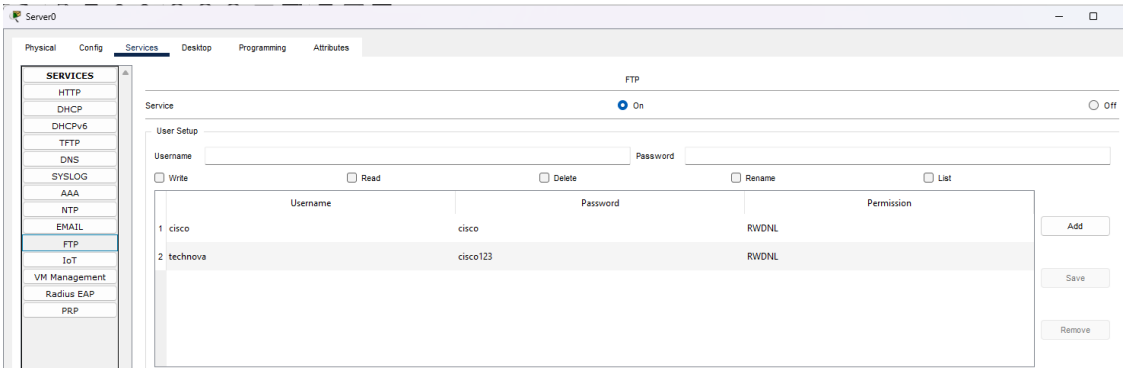


Figure 13. FTP service on, with the technova account added.

Testing before the ACL (steps 7 to 9)

Step 7: Test connectivity between departments

Before adding any restriction, I confirmed the basic network worked, so that any later failure could be traced to the ACL alone. From the Command Prompt on a PC in each department, I ran:

Test	Command	Result
Sales to IT	ping 192.168.10.5	Success
Sales to HR	ping 192.168.30.5	Success
HR to IT	ping 192.168.10.5	Success
HR to Sales	ping 192.168.20.5	Success

No screenshots were captured for these four pings.

Step 8: Test the HTTP server

From a PC outside the IT subnet, I opened Desktop > Web Browser and went to http://192.168.10.2. The NetCore Systems Ltd. page loaded, which shows that HTTP is on and that routing between departments works.

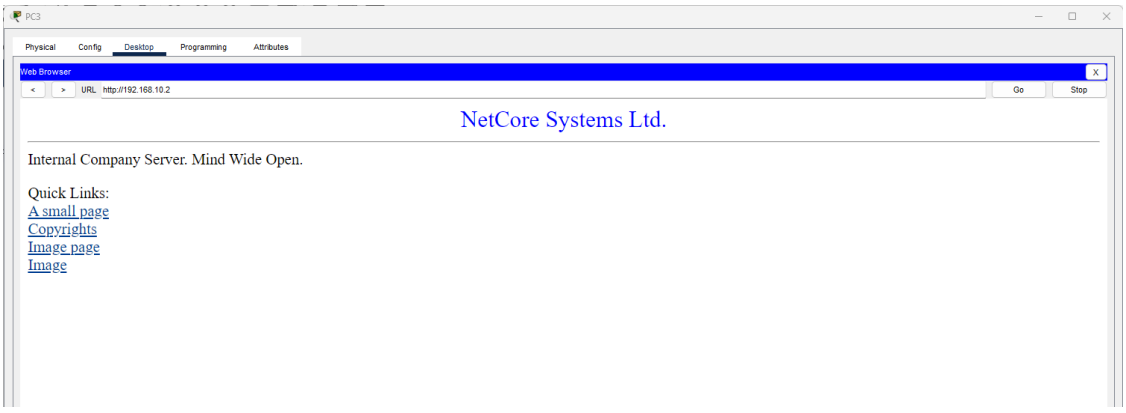


Figure 14. The company page loading on PC3 (HR) from 192.168.10.2.

Step 9: Test FTP

From PC5 in Sales, I ran ftp 192.168.10.2 in the Command Prompt and signed in as technova. The server replied, “230 Logged in”.

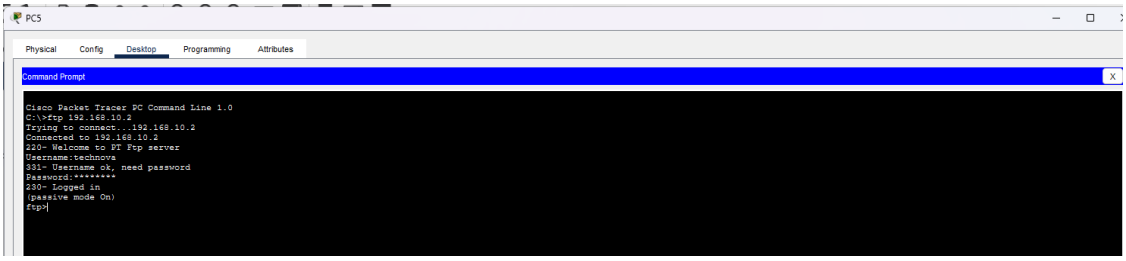


Figure 15. FTP login from PC5 (Sales) to the IT server.

Securing the network with an ACL (steps 10 to 12)

Step 10: Configure the ACL

The requirement is narrow: Sales must not be able to ping IT but must keep using the server's web and FTP services. Ping uses ICMP, while HTTP and FTP use TCP. I therefore used an extended ACL, which can filter by protocol, to deny only ICMP from the Sales subnet to the IT subnet and permit everything else.

```
enable
configure terminal
ip access-list extended Sales-ACL
 deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 permit ip any any
 exit
interface gigabitEthernet 0/1
 ip access-group Sales-ACL in
 exit
end
```

I applied the ACL inbound on G0/1, the Sales-facing interface. This filters Sales traffic as it enters the router, as close to the source as possible, and leaves HR and IT traffic untouched. The final "permit ip any any" is needed because every ACL ends with an implicit deny. No screenshot was captured of the commands being entered. Figure 21 shows the resulting ACL on the router.

Step 11: Test the ACL

I repeated the earlier tests with the ACL in place.

Sales to IT ping: blocked. From PC4, ping 192.168.10.5 returned "Destination host unreachable" from 192.168.20.1 with 100% loss. The reply comes from the router's Sales interface, which shows the router itself is refusing the traffic.

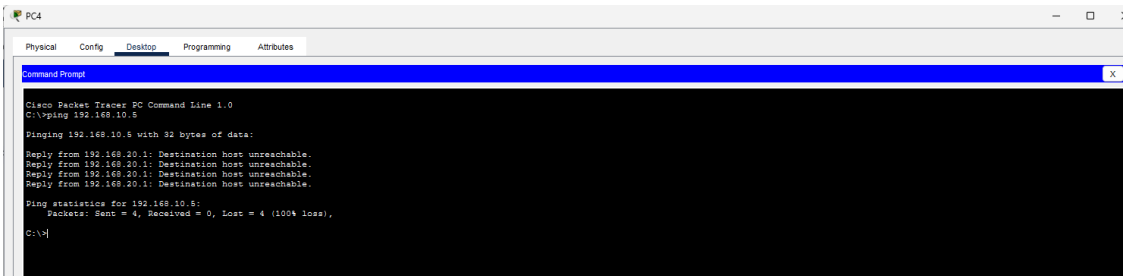


Figure 16. Sales (PC4) can no longer ping IT.

HR to IT ping: allowed. From PC2, ping 192.168.10.5 succeeded with 0% loss. HR is not affected by the ACL.

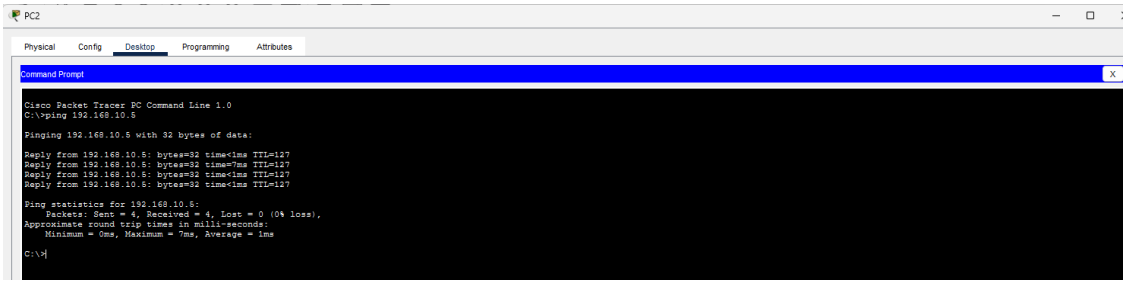


Figure 17. HR (PC2) can still ping IT.

HR to Sales ping: allowed. From PC3, ping 192.168.20.5 succeeded with 0% loss.

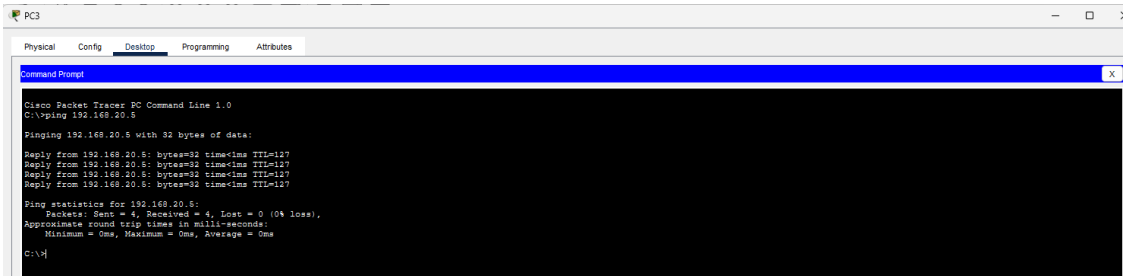


Figure 18. HR (PC3) can still ping Sales.

Sales to HTTP server: allowed. From PC5, http://192.168.10.2 still loads the company page.

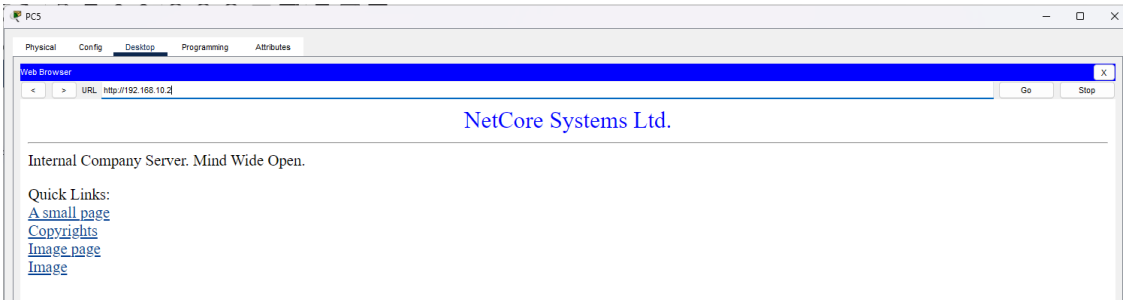


Figure 19. Sales (PC5) can still open the web page on the IT server.

Sales to FTP server: allowed. From a Sales PC, ftp 192.168.10.2 still connects and logs in as technova.



Figure 20. Sales can still log in to FTP on the IT server.

Step 12: Verify the ACL on the router

I ran show access-lists and show ip interface g0/1 on the router. The output shows both ACL entries, with 4 matches on the deny line, one for each blocked ping packet in Figure 16. It also confirms that the inbound access list on G0/1 is Sales-ACL and that the helper address is 192.168.10.2.

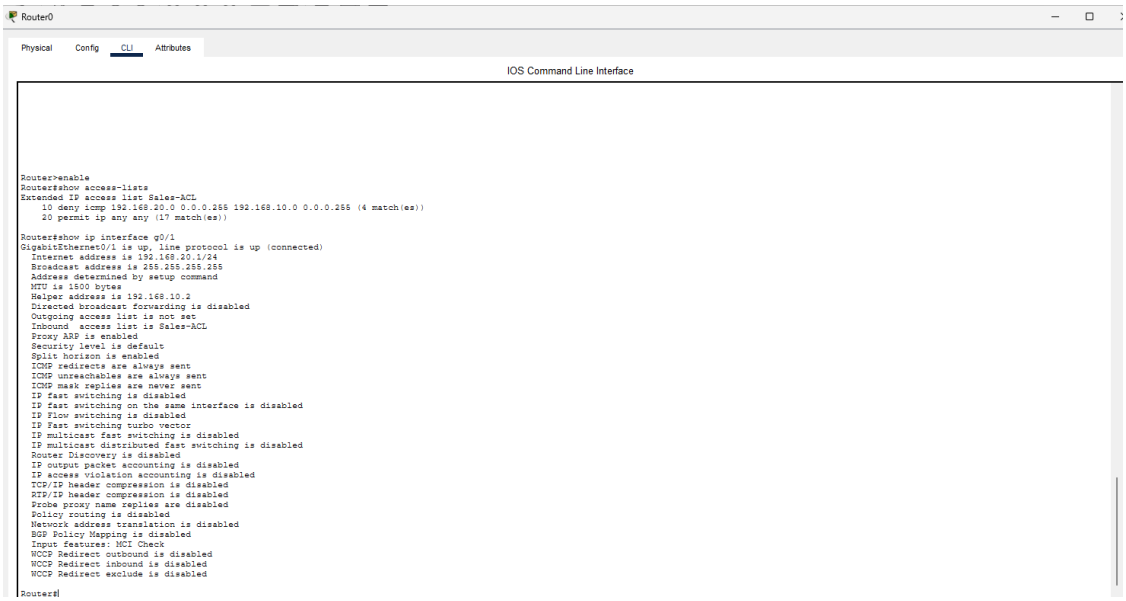


Figure 21. Sales-ACL with match counters, applied inbound on GigabitEthernet0/1.

Results

Every management requirement was met and verified.

Requirement	Outcome	Evidence
Separate segment per department	Three /24 subnets on three router interfaces	Figures 1 to 4
IP addresses assigned appropriately	All six PCs addressed by DHCP, server static	Figures 5 to 12
Central HTTP and FTP services	Both running on the IT server and reachable	Figures 13 to 15
Sales to IT ping blocked	Blocked by Sales-ACL	Figures 16 and 21
Sales keeps HTTP and FTP access	Both still work after the ACL	Figures 19 and 20
HR to IT and HR to Sales still work	Both pings succeed	Figures 17 and 18

The network now gives NetCore Systems Ltd. a structured base that can grow. A new department needs only another subnet, a router interface, and a DHCP pool, and further restrictions can be added as extra lines in the ACL.